pedk.ai - AI-native operational reconciliation engine

Version 1.0

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Pedk.ai — Product Specification Document

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> [!NOTE] This document was produced following the **Erdos AI Enterprise AI Deployment Methodology** — a structured, persona-driven approach that routes analysis through three lenses (Product, Business Case, Skills) to produce a comprehensive product specification. Superseded documents are archived in `docs/archive/`.

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1. Problem Statement

> **Erdos Template:** We need to [AI capability] in order to [business outcome] for [affected users], within the constraints of [limits], measured by [metric], owned by [accountable role].

We need to continuously reconcile the physical, electronic reality of large enterprise network infrastructure against its documented, bureaucratic intent (CMDBs, change tickets, architecture diagrams) **in order to** eliminate the "Dark Graph" — the mathematically provable divergence between what the network actually *is* and what the organisation *thinks* it is — **for** NOC engineers, IT operations leaders, and service assurance teams at Tier-1 telecom operators (Vodafone/Jio/Verizon scale), **within the constraints of** multi-vendor environments (Ericsson, Nokia, Huawei), TM Forum Open APIs (TMF628, TMF642), GDPR/ICO data protection requirements, OFCOM regulatory obligations, and zero-disruption deployment mandates, **measured by** CMDB accuracy improvement, mean time to resolve (MTTR) reduction, undocumented dependency discovery rate, and operational efficiency improvement, **owned by** the VP of IT Operations / Head of Service Management.

Why This Problem Exists

Every global enterprise is fundamentally blind to its own nervous system. The physical reality of infrastructure (telemetry, logs, packet flows) is structurally out of sync with the bureaucratic intent (static CMDBs, forgotten change tickets, outdated architecture diagrams). This divergence has four root causes:

Cause	Description	Data Availability
Intrusion & Anomaly	Malicious actors traverse infrastructure in ways no CMDB records	HIGH (CasinoLimit dataset)
Tribal Knowledge	Engineers maintain the real state in their heads; CMDBs capture the minimum	LOW (requires ticket corpus)
Secure Infrastructure	CyberArk-protected, air-gapped assets invisible to telemetry	NONE (synthetic simulation)
Multi-Party Handover	Decades of site deployments across dozens of contractors, each with independent identifiers	NONE (synthetic simulation)

Why Now

- Post-5G SA deployment: 200–400 OSS/BSS tools, >1,200 integration points per Tier-1 operator

- OPEX growing 8–12% YoY; 35–50% consumed by maintenance and integration tax
- Incumbent AI Service Mapping (BMC/ServiceNow) relies on active discovery pinging — blind behind firewalls
- Pedk.ai finds what active scanning cannot: the dependencies that exist between firewalled, air-gapped, or policy-shielded infrastructure

The Bottom Line

Nobody in the C-suite wakes up worrying about CMDB accuracy. But they do worry about what CMDB inaccuracy *causes*. Pedk.ai translates technical reconciliation into four hard executive value levers:

Executive Pain Point	Business Impact	Pedk.ai's Answer
The "Integration Tax" (OPEX)	Tier-1 operators spend millions annually on L2/L3 engineers blindly hunting for root causes because their maps are wrong. This is dead OPEX.	By eliminating the Dark Graph, troubleshooting goes from "hunting in the dark" to "following a lit path", structurally lowering MTTR and OPEX.
Silent Revenue Leakage	Sleeping cells report as "healthy" but serve zero traffic. Customers churn; SLAs are breached; revenue leaks silently.	Pedk.ai finds the failures that traditional element managers and alarms miss entirely, safeguarding top-line revenue.
Headline / Regulatory Risk	Major outages (like 999/911 failures) happen because an undocumented dependency cascaded unexpectedly. Fines are massive; brand damage is worse.	Continuous, automatic reconciliation makes network brittleness visible *before* the critical collapse occurs.
Phantom CapEx / Licences	Decommissioned but fully-licenced instances (Phantom Nodes) cost real money in software renewals and vendor maintenance.	Pedk.ai identifies assets generating zero telemetry for months, enabling immediate licence clawbacks and hard savings.

2. Product Vision & Value Proposition

Vision

Pedk.ai is an **AI-Native Operational Reconciliation Engine** — the intelligence fabric that connects existing enterprise systems (ITSM, BSS/OSS, vendor NMS) to translate raw electronic chaos into documented reality.

> [!IMPORTANT] **Pedk.ai does not replace Ericsson, Nokia, ServiceNow, or Amdocs.** Element managers exist for critical reasons: real-time control, vendor-specific configuration, regulatory compliance, and operational safety. Pedk.ai sits *above* these systems as a reconciliation layer. It observes, correlates, and keeps the business aware of the continuous churn between what element managers report and what the CMDB records. Pedk.ai augments — it never displaces.

The Dark Graph

The Dark Graph is any element of the real network topology that diverges from what the CMDB declares. It is Pedk.ai's moat and core value. It manifests at every level:

Divergence Type	Description	Example
Dark Nodes	Entities that physically exist but have no CMDB record	Elastically scaled VNF emitting telemetry; CMDB unaware
Phantom Nodes	CMDB records for entities that no longer exist	Decommissioned cell site still listed as "active"
Dark Edges	Undocumented connections between entities	SSH tunnel between two servers behind a firewall
Phantom Edges	CMDB-declared connections that carry no traffic	Firewall rule decommissioned but never removed from CMDB
Dark Attributes	Properties recorded incorrectly or not at all	Radio link parameters adjusted in the field, never updated

Divergence Type	Description	Example
Identity Mutations	Same logical function served by different physical entity	Emergency hardware swap: new serial number, same IP

Value Proposition

Value Lever	Mechanism	Impact
CMDB Reconciliation	Offline ingestion of historical ticket data → automated discovery of undocumented dependencies	300+ missing dependencies discovered in 48 hours
MTTR Reduction	Cross-domain causal root cause analysis via inferred topology	25–45% improvement in mean time to resolve
Operational Awareness	Continuous reconciliation of live element manager output against CMDB records	Real-time visibility into infrastructure churn that accumulates silently
Proactive CX	Churn-to-anomaly correlation triggers automated care workflows	At-risk customers identified before they complain

Go-to-Market: The “Read-Only Wedge”

Pedk.ai’s Day 1 entry point requires **zero write access** and **zero sensitive data exposure**. The customer provides three read-only datasets:

Data Source	Format	Purpose
CMDB snapshot	CSV/JSON export of Configuration Items and relationships	The “declared truth” — what the organisation thinks the network looks like

Data Source	Format	Purpose
Telemetry time-series	Historical PM counters, event logs, alarm history (12 months minimum)	The "physical truth" — what the network actually did
ITSM ticket archive	CSV export of incident and change tickets	The "human truth" — how engineers actually troubleshot and fixed problems

The Three-Way Cross-Correlation Pedk.ai's value comes from cross-correlating these three data sources against each other:

1. **Telemetry vs CMDB → Dark Nodes & Phantom CIs**: When telemetry arrives from entities that have no CMDB record, Pedk.ai identifies Dark Nodes. When CMDB CIs generate zero telemetry for months, Pedk.ai flags them as Phantom CI candidates — still consuming licence fees, wasting operational attention, but providing no service.
2. **Telemetry vs CMDB → Identity Mutations**: When telemetry shows a CI with a different serial number, MAC address, or hardware model than the CMDB records, Pedk.ai detects an undocumented hardware swap — the logical function persists but the physical entity has changed.
3. **Ticket patterns vs CMDB → Behavioural Dark Edges**: When engineers consistently access CIs A and B together during troubleshooting (evidence from ticket resolution notes and CI access logs), but the CMDB records no relationship between them, Pedk.ai infers an undocumented operational dependency. These fragments also feed the Abeyance Memory (§4) for future cross-modal matching.
4. **Telemetry patterns → Evidence Fusion**: The evidence fusion engine (§5) populates the vector store with relationship hypotheses. Algorithms score the confidence of each discovered divergence. Over time, corroborated hypotheses graduate through the lifecycle (`CANDIDATE → CORROBORATED → ACCEPTED`).

Delivery: Within 48 hours, Pedk.ai hands back a Divergence Report. This is a 3-month sales cycle. The customer risks nothing — they share only historical operational data, not revenue, not customer PII, not billing records. Trust is earned through demonstrated value before any deeper integration is discussed.

3. Architecture Overview

Pedk.ai is a 5-layer intelligence architecture:

```
+-------------------------------------------------------------------+
|                      PEDKAI CONTROL PLANE                         |
+-------------------------------------------------------------------+
|  Layer 5: Automation & Actuation                                  |
|  - Zero-touch ITSM ticket creation & routing                      |
|  - Granular config changes via Vendor APIs (Ansible/NetConf)      |
|  - Gated autonomous actions with 7 safety gates                   |
|  - Rule: No autonomous action without explainability              |
|  - OPERATOR ALWAYS IN CONTROL (see S7)                            |
+-------------------------------------------------------------------+
|  Layer 4: Decision & Policy Engine ("The Constitution")           |
|  - YAML-based declarative business rules                          |
|  - Risk-aware decision making bound by SLA constraints            |
|  - Decision Trace capture, pattern matching, outcome learning     |
+-------------------------------------------------------------------+
|  Layer 3: Intelligence Engines                                    |
|  - Anomaly Detection (Z-score, statistical)                       |
|  - Root Cause Analysis (graph-based causal inference -- S8)       |
|  - LLM Reasoning (vendor-neutral -- see S15)                      |
|  - Decision Similarity Search (vector embeddings)                 |
|  - Evidence Fusion (pluggable methodology -- see S5)              |
+-------------------------------------------------------------------+
|  Layer 2: Living Context Graph (Decision Memory)                  |
|  - Decision Traces (why decisions were made, not just what)       |
|  - Evidence Snapshots & Constraint History                        |
|  - ABEYANCE MEMORY -- Pedk.ai's core differentiator (see S4)      |
|  - Topological Ghost Masks (suppress alarms during maintenance)   |
+-------------------------------------------------------------------+
|  Layer 1: Omniscient Data Fabric                                  |
|  - Physical Reality: Telemetry (Kafka), PM counters, logs         |
|  - Bureaucratic Intent: ITSM tickets, CMDB exports, changes       |
|  - Multi-vendor ingestion: Ericsson (XML), Nokia (JSON)           |
|  NOTE: Pedk.ai OBSERVES element manager output.                   |
|  It does not replace or bypass element managers.                  |
+-------------------------------------------------------------------+
```

Reconciliation Engine

The core mathematical engine uses a **single hypothesis lifecycle** that handles two orthogonal axes:

- **What axis** (topology element): Dark nodes, phantom nodes, identity mutations, dark edges, phantom edges, dark attributes — all pass through `CANDIDATE → CORROB-`

`ORATED → ACCEPTED → INTEGRATED`

- **Why axis** (causal type): Intrusion, tribal knowledge, secure infrastructure, multi-party handover — all contribute evidence weighted by source type

The evidence fusion methodology (currently Noisy-OR) is **pluggable** — see §5 for alternatives and per-customer selection rationale.

4. Abeyance Memory — Pedk.ai's Core Differentiator

What It Is

Abeyance Memory is Pedk.ai's proprietary capability to hold **disconnected, unresolved technical facts** in a latent semantic buffer — indefinitely — until the missing contextual link appears. Unlike conventional monitoring systems that process events in real-time and discard unresolvable data, Pedk.ai *remembers* fragments that don't yet make sense.

How It Works

1. **Capture**: During routine ticket processing, a NOC engineer pastes a CLI output into a resolution note. The output references an IP address and a timeout error. Pedk.ai ingests this fragment even though no current alarm or CMDB record matches it.
2. **Hold in Abeyance**: The fragment is embedded into the vector store with its full semantic context (timestamp, engineer ID, ticket ID, affected CI, raw text). It sits in the Abeyance Memory — not discarded, not acted upon — waiting.
3. **Snap**: Three weeks later, a new alarm fires from a previously-unknown entity on the same subnet. The anomaly detector flags it. Pedk.ai's similarity search against the Abeyance Memory hits the stored fragment at 0.92 similarity. The two pieces of evidence — a human observation from 3 weeks ago and a machine observation from today — are **snapped together** into a corroborated hypothesis.
4. **Resolve**: The hypothesis graduates from `CANDIDATE` to `CORROBORATED`. Pedk.ai now has a Dark Edge linking two entities that no active scanner could have discovered — because the evidence existed in two different time horizons and two different data modalities (human text + machine telemetry).

Why It Matters

- **Active discovery scanners** (BMC, ServiceNow) operate in real-time only. If the signal isn't present when the scan runs, it's missed forever.
- **Rule-based correlation engines** require explicit rules to match events. If no rule exists for the specific pattern, it's ignored.

- **Abeyance Memory** is patient. It accumulates evidence across weeks, months, and even years. It doesn't need an explicit rule — it uses semantic similarity to find connections that no human analyst could spot across thousands of tickets and millions of events.

The Competitive Moat

No competing product holds unresolved evidence in persistent semantic storage with the intent to match it later. This makes Abeyance Memory Pedk.ai's most defensible technical differentiator. It transforms "noise" into "future intelligence."

Engineering Maturity

Aspect	Status	Gap
Vector storage of fragments	✅ Implemented (pgvector)	—
Semantic similarity snapping	✅ Implemented	Threshold tuning needed per deployment
Multi-modal matching (text + telemetry)	⚠️ Partial	Need structured telemetry-to-text alignment
Long-horizon retrieval (>30 days)	⚠️ Partial	Cold storage retrieval pipeline incomplete
Abeyance decay and relevance scoring	❌ Not implemented	Stale fragments need TTL with relevance weighting

5. Evidence Fusion Methodology

Current Implementation: Noisy-OR

Pedk.ai currently uses the **Noisy-OR** probabilistic model for combining evidence from independent sources. Each evidence source contributes an independent probability of causing a hypothesis to be true, and the combined confidence follows the formula:

$$P(\text{hypothesis} \mid \text{evidence}_1, \text{evidence}_2, \ldots, \text{evidence}_n) = 1 - \prod_i (1 - p_i)$$

This works well when evidence sources are genuinely independent and their influences are additive.

Critical Assessment

Noisy-OR is a sound starting point, but **no single fusion methodology fits all customers or all evidence types**. Different operators have different evidence profiles, and the mathematical model should be selected to match the customer's data landscape.

Alternative Methodologies Under Consideration

Methodology	How It Works	Best For	Limitations	Pedk.ai Fit
Noisy-OR (current)	Each cause independently contributes to hypothesis truth; probabilities combine multiplicatively	Multiple independent telemetry signals confirming an edge or node	Assumes independence; struggles with correlated evidence; no concept of "ignorance"	✅ Good for Type 1 (intrusion/anomaly) with rich, independent telemetry
Dempster-Shafer Theory	Assigns belief to *sets* of hypotheses, explicitly representing ignorance; combines via Dempster's Rule	Sparse evidence environments where "I don't know" is a valid and important state	Can produce counter-intuitive results when evidence is highly conflicting; computationally intensive	✅ Strong fit for Type 3 (secure infrastructure) and Type 4 (multi-party handover) where evidence is sparse and partial
Fuzzy Logic	Assigns degrees of membership (0–1) to linguistic categories; combines via t-norms	Operator-derived qualitative assessments ("high confidence", "probable cause")	No statistical foundation; subjective membership functions; poor for learning from data	⚠️ Useful for policy engine rules but not for evidence fusion

Methodology	How It Works	Best For	Limitations	Pedk.ai Fit
Noisy-MAX	Generalises Noisy-OR to multivalued ordinal variables; effect determined by strongest cause	Severity classification where multiple anomalies compete to determine the dominant failure mode	Requires ordinal ordering of states; less flexible than Noisy-OR for binary hypotheses	⚠️ Potential fit for alarm severity arbitration
Bayesian Network (full)	Complete directed acyclic graph with learned conditional probability tables	Rich causal structure with known dependencies between evidence types	Exponential CPT growth; requires training data for structure learning; computationally expensive	🔮 Future: when sufficient production data is available for structure learning

Recommended Strategy: Per-Customer Selection

The evidence fusion methodology should be **configurable per deployment**, matched to the customer's evidence profile:

Customer Profile	Primary Evidence	Recommended Methodology
Operator with rich telemetry (Kafka streams, gNMI)	Dense, real-time, independent signals	**Noisy-OR** (current)
Operator with sparse, policy-gated data	Partial ticket references, fragmentary evidence	**Dempster-Shafer**
Operator with qualitative NOC assessments	Operator judgment, experience-based ratings	**Fuzzy Logic overlay** on Noisy-OR
Operator with extensive historical ticket corpus	Behavioural patterns across years of ITSM data	**Full Bayesian Network** (with structure learning)

> [!IMPORTANT] **Product roadmap item**: Implement a `FusionMethodologyFactory` that selects the appropriate evidence fusion engine at deployment configuration time. V1 ships with Noisy-OR. V2 adds Dempster-Shafer for sparse-evidence deployments. See §17 Task Backlog for engineering tasks.

6. Feature Inventory

> [!WARNING] **Engineering Maturity Notice**: The features listed below represent the product's designed scope. Many are implemented at proof-of-concept or early-prototype maturity. Significant engineering work is required to bring them to production-ready quality. Each feature includes a maturity status. Items marked ⚠️ or 🔨 require focused engineering before they can be demonstrated to customers with confidence.

Feature 1: Dark Graph Reconciliation (The Wedge)

Why this is Feature #1: This is Pedk.ai's most immediately deliverable capability. It can be demonstrated using **read-only historical data** with **zero infrastructure changes** on the customer side. It is the proof point that earns trust.

Capability	Description	Maturity
Divergence Report	Machine-generated report of all dark nodes, phantom nodes, identity mutations, dark/phantom edges discovered from historical data	🔨 In Progress
Datagerry CMDB sync adapter	Periodic snapshots of CMDB state into Pedk.ai's internal representation	🔨 In Progress
CasinoLimit telemetry parser	Three telemetry streams: network flows, syscalls, MITRE labels	🔨 In Progress
Topological Ghost Masks	Suppresses alarms during planned maintenance by cross-referencing change schedules	🔨 In Progress

Capability	Description	Maturity
Behavioural Dark Graph	Infers operational dependencies from how engineers actually troubleshoot (CI access sequences)	📋 Planned
Multi-party identity resolution	Cross-lifecycle identifier clustering from unstructured deployment documents	📋 Planned

Day 1 delivery model: Customer provides read-only access to three historical datasets: CMDB snapshot, telemetry time-series (12 months), and ITSM ticket archive (see Go-to-Market: The Read-Only Wedge for the three-way cross-correlation mechanism). Within 48 hours, Pedk.ai delivers a Divergence Report identifying Dark Nodes, Phantom CIs, Identity Mutations, and Behavioural Dark Edges. Value demonstrated before any production integration.

Feature 2: Autonomous Network Operations (ANOps)

Capability	Description	Maturity
Multi-service anomaly detection	Z-score statistical analysis across mobile, voice (VoLTE), SMS, landline	✅ Implemented
Graph-based root cause analysis	Recursive graph traversal mapping anomalies to upstream dependencies	✅ Implemented
LLM-powered SITREPs	Vendor-neutral LLM generates actionable explanations for NOC engineers	✅ Implemented
Sleeping cell detection	Multi-method silent failure detection (see below)	⚠️ Code exists but not wired into scheduler
Congestion management	PRB utilisation vs latency correlation, DSS activation recommendations	✅ Implemented
Emergency compliance	999/911 dial-out blockage detection and life-critical traffic override	✅ Implemented

Sleeping Cell Detection — Technical Methodology Sleeping cells are the hardest network failures to detect because **they generate no alarms**. The cell reports "green" to the element manager — but it serves zero users. The alarm gap itself is the diagnostic signal. Pedk.ai uses six complementary detection methods:

#	Method	How It Works	What It Catches
1	**Zero-User Baseline Comparison**	Compares active user count against historical baseline for same cell, same time-of-day, same day-of-week. If users drop to zero when the baseline predicts >50, the cell is flagged.	Complete silent failures — cell is up but not serving any traffic
2	**Neighbour-Reference Z-Score**	For each cell, calculates Z-score of key KPIs (throughput, RRC connections, PRB utilisation) relative to its geographic and topological neighbours. Healthy neighbours with similar coverage should show similar traffic patterns.	Partial degradation — cell serves some traffic but significantly less than neighbours, indicating it's silently shedding load
3	**KPI Correlation Breakpoint Detection**	Monitors the correlation between KPI pairs that should move together (e.g., number of users ↔ downlink throughput ↔ PRB utilisation). When correlations that have been stable for weeks suddenly break (users rise but throughput doesn't), regression analysis flags the anomaly.	Subtle degradation — cell appears operational but internal performance chains are broken

#	Method	How It Works	What It Catches
4	**Deep Autoencoder Reconstruction Error**	Trains an autoencoder on "normal" multi-KPI vectors per cell type. At detection time, measures reconstruction error. High reconstruction error indicates KPI patterns that don't conform to learned normal behaviour, even when no individual KPI breaches a threshold.	Complex multi-dimensional failures that no single-KPI threshold would catch
5	**MDT-Based Triangulation** (where available)	Uses Minimisation of Drive Tests (MDT) reports from UE devices. If devices in a cell's coverage area consistently report poor RSRP/SINR while the cell itself reports normal operation, the cell is failing to provide service despite being "up."	Radio coverage failures where the cell is operational but its signal is impaired
6	**Traffic Handover Asymmetry**	Analyses handover patterns. If neighbouring cells consistently hand *in* to the suspect cell but it never hands *out*, or if its handover success rate drops while neighbours remain stable, the cell may be accepting connections but failing to maintain them.	Cells that accept incoming handovers but silently drop users after connection

Design principle: Sleeping cells produce **no alarms** — the alarm gap is the ground-truth signal. The synthetic data from Sleeping-Cell-KPI-Data is specifically designed to generate sleeping cell scenarios with zero alarms, forcing Pedk.ai to detect them through KPI pattern analysis.

Feature 3: AI-Driven Capacity Planning

Capability	Description	Maturity
Data-driven densification engine	Queries real KPI hotspots (congestion >85%) from time-series store	⚠️ Prototype
CapEx optimisation	Greedy ROI-based site placement selection within budget constraints	⚠️ Prototype
Regional densification dashboard	Integrated visualisation in NOC UI	⚠️ Prototype

Feature 4: Customer Experience Intelligence

Capability	Description	Maturity
Per-subscriber network QoE scoring	Multi-dimensional quality score per customer based on cells they connect to	⚠️ Prototype
Churn-to-anomaly correlation	Maps network degradation events to at-risk subscribers	⚠️ Prototype
Proactive care automation	Automated care triggers and notification logging	⚠️ Prototype

Customer Experience Intelligence — Technical Methodology The core technical challenge is connecting **network-layer events** (which Pedk.ai sees) to **individual customer impact** (which matters to the business). The pipeline works as follows:

Step 1: Per-Subscriber Network QoE Scoring

Each subscriber connects to specific cells at specific times. By joining subscriber connection records (from BSS data, when available in future phases; or from anonymised network attachment records for Day 1) with cell-level KPIs, Pedk.ai constructs a **per-subscriber Quality of Experience (QoE) score**:

QoE Dimension	KPI Source	Weight
Coverage reliability	RSRP/SINR from serving cell(s)	30%
Data experience	DL/UL throughput, latency	25%
Voice quality	VoLTE MOS, call setup success rate	20%
Service availability	Sleeping cell exposure, outage duration in coverage area	15%
Stability	Handover failure rate, RRC re-establishments	10%

The score is calculated on a rolling 7-day window. A subscriber whose primary cells are degrading will see their QoE score decline even before they notice or complain.

Step 2: Churn Risk Correlation

When QoE scores decline for a cluster of subscribers in a geographic area, Pedk.ai correlates this with: - Historical churn patterns: subscribers with sustained QoE < threshold for >14 days have statistically higher churn rates - SLA tier: Enterprise and Gold subscribers have lower tolerance thresholds - Competitive coverage: areas where competitor coverage is strong amplify churn risk

Step 3: Proactive Care Triggers

When the correlation engine identifies at-risk subscribers, Pedk.ai generates: - **NOC SITREP**: identifying the network root cause (e.g., degraded cell, sleeping cell in coverage area) - **CX Alert**: identifying affected subscribers by segment, SLA tier, and estimated revenue at risk - **Care Recommendation**: suggesting proactive actions (e.g., SMS apology, data credit, priority repair scheduling)

Key insight: Pedk.ai's value is not the churn prediction model itself (every telco has one). The value is **connecting the network root cause to the customer impact** — something no existing system does because network data and customer data live in different silos. Pedk.ai's Dark Graph reconciliation is what bridges them.

Strategic Capabilities (Future Roadmap)

Capability	Description	Maturity
BSS Data Layer	Revenue and billing context for risk assessment	🔮 Future roadmap — requires earned customer trust (see §13)
Policy Engine (“Constitution”)	YAML-based declarative business rules for SLA enforcement	✅ Implemented
Semantic Context Graph	Recursive reasoning across multi-hop relationships	⚠️ Prototype
Closed-Loop RL Evaluator	Reinforcement learning from incident closure outcomes	⚠️ Prototype

7. Operator Feedback & Human-in-the-Loop

> [!IMPORTANT] **Pedk.ai can be autonomous, but it does not have to be.** Autonomy is a spectrum. Every customer chooses their comfort level. The operator is always in control.

The Autonomy Spectrum

Level	Name	Pedk.ai’s Role	Operator’s Role
0	**Advisory Only**	Generates SITREPs. Recommends actions. Takes zero action.	Full manual control. Uses Pedk.ai as a diagnostic assistant.
1	**Assisted**	Creates draft tickets, pre-populates fields.	Reviews, approves, and dispatches every ticket.
2	**Supervised**	Executes routine actions (e.g., ticket creation) with operator override window.	Monitors, can veto any action within a configurable window.
3	**Gated Autonomous**	Executes pre-approved action types (e.g., alarm acknowledgement) with 7 safety gates.	Reviews safety gate audit trail. Can invoke kill-switch at any time.

Default deployment: Level 0. The customer advances only when they choose to.

How Operator Feedback Works

Pedk.ai captures operator feedback through **three channels**, ranked by signal quality:

Channel	Signal Type	Quality	Implementation
Behavioural Observation	What the operator *actually does* after receiving a SITREP — which tickets they open, which CIs they check, which resolution actions they take, how they modify Pedk.ai's draft tickets	🟢 Highest	Ingests operator actions from ITSM (ticket modifications, CI interactions, resolution codes) and correlates with Pedk.ai's recommendations
Structured Assessment	Operator rates decision quality on a scale (e.g., accuracy 1–5, relevance 1–5, actionability 1–5) with optional freeform notes	🟡 Medium	Multi-dimensional assessment forms integrated into NOC dashboard
Quick Signal	Thumbs up / thumbs down on a specific recommendation	🔴 Lowest	Simple binary feedback; useful for aggregate trends but insufficient for technical nuance

> [!IMPORTANT] **Thumbs up/down alone is a poor quality signal for a technically nuanced product.** A "thumbs down" tells Pedk.ai that something was wrong but not *what* was wrong — was the root cause identification incorrect? Was the recommendation impractical? Was the severity miscalibrated? Pedk.ai prioritises behavioural observation (what did the operator *actually do*?) because actions speak louder than clicks.

Feedback Loop Mechanics

1. Pedk.ai generates a SITREP and recommended action
2. Operator reads the SITREP and takes action (or ignores it)
3. Pedk.ai observes the operator's actual response (via ITSM integration)
4. The delta between "what Pedk.ai recommended" and "what the operator actually did" is the **learning signal**

5. This delta updates the Decision Memory's relevance weighting and the similarity search's ranking function
6. Over time, Pedk.ai's recommendations converge toward the patterns that operators actually follow

Engineering Maturity

Aspect	Status	Gap
Thumbs up/down binary feedback	✅ Implemented	—
Multi-operator feedback aggregation	✅ Implemented (junction table)	Need anti-gaming safeguards
Behavioural observation pipeline	❌ Not implemented	Critical gap — highest value signal channel
Structured multi-dimensional assessment	❌ Not implemented	Need NOC dashboard integration
ITSM action ingestion (ServiceNow/Remedy)	❌ Not implemented	Depends on customer ITSM platform
Decision recommendation → outcome tracking	⚠️ Partial	RL evaluator exists but not fully wired

8. Causal Inference Methodology

Current Implementation: Granger Causality

Pedk.ai uses **Granger Causality** for time-series causal inference — testing whether past values of metric X improve the forecast of metric Y beyond Y's own history. Implementation includes ADF stationarity tests, automatic differencing, and dynamic candidate metric discovery.

Critical Assessment

Granger Causality is appropriate for V1 but has known limitations that will constrain Pedk.ai's effectiveness as it scales to more complex network topologies:

Limitation	Impact on Pedk.ai
Detects only **linear** relationships	Non-linear cascading failures (e.g., exponential congestion buildup) will be missed
Requires **stationarity**	Real network KPIs are often non-stationary (trending traffic growth, seasonal patterns)
Bivariate by default	Real network failures involve multi-variable cascading; pairwise testing misses confounders
Omitted variable bias	If the true cause isn't in the candidate set, Granger will attribute causality to a correlated proxy

Alternative Methodologies

Method	How It Works	Strengths	Limitations	Pedk.ai Roadmap
Granger Causality (current)	VAR model: does X's past predict Y's future?	Simple, fast, well-understood	Linear only, bivariate, stationarity required	✅ V1
Transfer Entropy	Information-theoretic: how much does knowing X's past reduce uncertainty in Y's future?	Captures **non-linear** dependencies; model-free	Computationally expensive; sensitive to data volume and discretisation	🔮 V2
PCMCI (Peter Clark – Momentary Conditional Independence)	Constraint-based causal graph discovery with conditional independence tests adapted for time series	Discovers full **causal graph** (not just pairwise); handles multivariate; controls for confounders	Computationally intensive; requires large sample sizes	🔮 V2

Method	How It Works	Strengths	Limitations	Pedk.ai Roadmap
Convergent Cross Mapping (CCM)	Dynamic systems theory: if X causes Y, then Y's history contains information about X (attractor reconstruction)	Works for **non-linear, non-separable** systems; identifies weak causality Granger misses	Requires long time series; embedding parameter sensitivity; may show false bidirectional causality under strong forcing	🔮 V3
Structural VAR (SVAR)	VAR with contemporaneous effect identification via restrictions	Models simultaneous causation (events happening at same time step)	Requires domain knowledge for identifying restrictions; still linear	🔮 V2

Recommended Roadmap

- **V1** (current): Granger Causality with ADF stationarity tests
- **V2**: Add Transfer Entropy for non-linear detection and PCMCI for full causal graph discovery
- **V3**: Add CCM for complex coupled systems (relevant for RAN-Transport-Core cascading failures)
- **Per-customer selection**: Like evidence fusion (§5), causal methodology should be configurable per deployment based on data characteristics

9. Data Model & Multi-Tenant Design

Multi-Tenancy Architecture

Pedk.ai follows the **industry-standard SaaS multi-tenancy model**:

- **All database tables carry a `tenant_id` column** — every query is scoped by tenant
- **Tenant data boundary is sacrosanct** — no cross-tenant data access, no cross-tenant analytics, no exceptions
- **Each tenant has isolated configuration**, security policies, and data retention rules
- Managed Service Providers (MSPs) hosting Pedk.ai for multiple operators get per-tenant dashboards, never aggregated views

Core Data Entities

ORM Model	Purpose
`NetworkEntityORM`	Network topology nodes (cells, routers, VNFs)
`EntityRelationshipORM`	Topology edges (from_entity → to_entity)
`KPIMetricORM` / `KpiSampleORM`	Time-series PM counters
`DecisionTraceORM`	Context Graph decision memory (with `embedding_provider`, `embedding_model`)
`DecisionFeedbackORM`	Multi-operator feedback (junction table, per-operator voting)
`CustomerORM` / `BillingAccountORM`	BSS data layer (future roadmap)
`IncidentORM`	Incident lifecycle management

Decision Trace Structure

Each decision captured by Pedk.ai includes context (alarm IDs, KPI snapshots, related tickets), binding constraints (SLA, maintenance windows), options considered with risk assessments, the tradeoff made, actions taken, outcomes measured, and learnings extracted. This is the "Decision Memory" — storing *why* decisions were made, not just *what* happened.

Storage Architecture

Tier	Purpose	Technology	Retention
Hot	Anomaly detection, real-time dashboard	Redis / TimescaleDB Hypertable	24–48 hours
Warm	RCA, graph traversal, decision memory	PostgreSQL (JSONB + pgvector)	30–90 days
Cold	Model training, historic audit, compliance	S3 / Parquet / Apache Iceberg	1–7 years

10. Synthetic Data & Testing Infrastructure

> [!IMPORTANT] Pedk.ai's synthetic data is generated by the **Sleeping-Cell-KPI-Data** project (`/Users/himanshu/Projects/Sleeping-Cell-KPI-Data`) — producing converged telecom operator datasets for Dark Graph training and ML model development.

Scale Parameters

The generator produces **~12.2 GB across 17 Parquet files**:

Parameter	Value
Sites	21,100 across 38 Indonesian provinces (3 timezones)
Logical cell-layers	~66,100 (LTE, NR-NSA EN-DC, NR-SA)
Network entities	~811,000 across 6 domains
Relationships	~1.97M with full cross-domain dependency chains
Hourly KPI rows	47.6M (30 days × 720 intervals × 66k cells)
Customers	1M sampled subscribers with SLA tier data
Vendors	Ericsson (55%) and Nokia (45%) naming layers

Synthetic Data Quality — The Achilles Heel

> [!CAUTION] **The realism of synthetic data is the single most critical risk to Pedk.ai's credibility.** A good engine trained on poorly constructed synthetic data will learn bad behaviours and draw inaccurate conclusions. When such a product encounters real-world data, it will act incorrectly. This must be prevented at all costs.

Current concerns:

Issue	Description	Impact	Required Action
UUID V4 overuse	Synthetic data uses UUID V4 for entity identifiers	Real operators use human-friendly IDs (e.g., `LTE-8842-A`, `SITE-NW-1847`, `WO-2024-NW-1847`)	Implement operator-realistic naming conventions with collision-safe generation
Scenario realism	Injected faults may not reflect real-world failure patterns	Model learns synthetic failure modes that don't occur in production	Validate scenario definitions against published post-incident reports from Tier-1 operators
Temporal realism	KPI time-series generated via AR(1) state machines	Real network KPIs have diurnal patterns, seasonal effects, and event-driven spikes that AR(1) may underrepresent	Implement empirical distribution sampling from reference datasets
Alarm correlation chains	Synthetic cross-domain cascades may oversimplify real propagation delays	Model may expect instantaneous cascading when real propagation has variable latency	Add configurable propagation delay profiles per domain boundary

Issue	Description	Impact	Required Action
CMDB degradation patterns	Synthetic divergence may not reflect actual CMDB decay rates	Dark Graph detection tuned to synthetic decay rates may miss slower real-world drift	Calibrate degradation rates against published CMDB audit statistics

Pedk.ai Integration Points

Pedk.ai Component	Fed By
`KPIMetricORM` / `KpiSampleORM`	Wide-format KPI Parquet files
`NetworkEntityORM`	`cmdb_declared_entities.parquet`
`EntityRelationshipORM`	`cmdb_declared_relationships.parqu`
`CustomerORM` / `BillingAccountORM`	`customers_bss.parquet` (future)
Alarm ingestion API (TMF642)	`events_alarms.parquet`
Dark Graph scoring	`divergence_manifest.parquet` + ground truth files

11. TMF Standards Compliance

TMF642 — Alarm Management API

- Standardised exposure of decision traces as network alarms
- Dual-correlation IDs: `external_correlation_id` (vendor NMS) and `internal_correlation_id` (Pedk.ai RCA)
- Alarm lifecycle: `raised` → `acknowledged` → `cleared`
- OAuth2 scope-based access enforcement

TMF628 — Performance Management API

- Standards-compliant KPI exposure (throughput, latency, PRB utilisation)

- PerformanceMeasurement and IndicatorSpec Pydantic models

Multi-Vendor Ingestion

Vendor	Format	Ingestion
Ericsson (ENM)	XML alarms	Kafka + REST, `pm` prefix PascalCase counters
Nokia (NetAct)	JSON alarms	High-volume Kafka, `VS.` dot-separated hierarchy
Generic SNMP	Standard traps	Kafka consumer

Alarm Normaliser: Pluggable strategy-based layer translating vendor payloads into unified internal signal format.

12. Deployment Model

Infrastructure

Component	Specification
API	FastAPI on Python 3.11+, multi-stage Docker container (non-root)
Database	PostgreSQL + JSONB + pgvector + TimescaleDB extension
Streaming	Apache Kafka
Container orchestration	Kubernetes with Helm charts
Secrets	Environment-injected; K8s Secrets template; TLS/SSL for DB

Deployment Modes

Mode	Purpose	Data Access
Offline PoC	Day 1 wedge: historical data analysis, Divergence Report generation	Read-only. Customer provides CMDB snapshot, telemetry time-series, and ITSM ticket archive. Zero production access required.
Shadow Mode	Parallel-run alongside existing tools; proves accuracy without taking control	Read-only production data feeds (Kafka tap, syslog mirror). No write access to any system.
Advisory Mode	Generates SITREPs and recommendations alongside L2 workflows	Read-only production feeds + advisory output to NOC dashboard. No automated actions.
Assisted Mode	Creates draft tickets, pre-populates fields; operator approves every action	Read + limited write (ITSM draft creation). Operator approval gating.
Production	Full deployment with operator-selected autonomy level (see §7)	Read + write per customer policy. All actions audited.

13. Security Model

Authentication & Authorisation

- JWT-based authentication with signature verification
- Hierarchical RBAC: Admin → Operator → Viewer
- OAuth2 scopes per API domain
- Functional `/api/v1/auth/token` endpoint

Data Protection

- Multi-tenant isolation: all queries scoped by `tenant_id` (§9)
- GDPR consent enforcement for proactive customer communications
- Data sovereignty guardrails: configurable data residency and egress controls
- Sensitive data scrubbing before external LLM provider calls

Day 1 Trust Architecture

> [!IMPORTANT] **BSS integration (revenue, billing, customer PII) is explicitly deferred to a future roadmap milestone.** Pedk.ai's Day 1 deployment accesses only operational data: CMDB snapshots, alarm histories, ticket exports, and PM counters. This is deliberate — asking for BSS data before earning trust is a recipe for rejection. Walk before we run.

The trust progression: 1. **Day 1**: Read-only CMDB + historical tickets → Divergence Report (zero risk) 2. **Month 3**: Read-only production telemetry feeds → Shadow Mode (zero risk) 3. **Month 6**: Advisory Mode with SITREP generation → Operator trust established 4. **Month 12+**: BSS integration discussion only after proven value delivery

Regulatory Compliance

Framework	Document	Status	Action Required
OFCOM	Pre-notification document	❌ Stub	Substantive rewrite required — must cover architecture, risk analysis, vendor compatibility
ICO	Data Protection Impact Assessment	❌ Stub	Full DPIA covering data flows, retention periods, consent mechanisms, international transfers
Safety	Autonomous Safety Whitepaper	❌ Stub	Architecture safety analysis, failure modes, testing methodology, rollback procedures
Autonomy	Status Report	❌ Stub	Operational status template for ongoing autonomous action reporting
Internal	ADR-001 (Autonomy Positioning)	✅ Complete	—
Internal	ADR-002 (Autonomous Execution Architecture)	✅ Complete	—

> [!CAUTION] **Regulatory documents are currently stubs.** This is a critical gap. No customer engagement with OFCOM-regulated operators can proceed until these documents contain substantive content. This is a top-priority task item.

14. Erdos Lens Analysis

This section applies the **Erdos AI Enterprise AI Deployment Methodology** three-lens analysis.

Pedk.ai's Committee Review Cycle

Before the Erdos analysis, it is worth noting the rigorous review process Pedk.ai has already undergone. The product was subjected to a **multi-phase committee review cycle** involving:

- **Forensic Technical Auditor** (Pass 1): Line-by-line codebase audit against roadmap intent
- **Architecture Viability Review** (Pass 2): Structural integrity, design patterns, scale readiness
- **Adversarial Review** (Pass 3): Red-team perspective on claims vs evidence
- **Five-member executive committee**: Ops Director, CEO, Strategist, Enterprise Architect, QA Director
- **Formal remediation cycle**: 46 tasks identified across 6 phases, tracked in `3Pass-ReviewOutcome_Roadmap_V3.yaml`

This review cycle resulted in the formal architecture review document (`ARCHITECTURE_S3_REMEDIATIO` which identified 6 Critical, 9 High, 10 Medium, and 4 Low findings — all with concrete remediation plans. The committee process itself is a product quality signal: Pedk.ai builds iteratively and subjects itself to structured criticism.

Lens 1 — Product Readiness

Dimension	Score	Rationale	Improvement Task
Problem & Value	4	Problem is specific, measurable (MTTR, CMDB accuracy), urgently felt. Stable over time. Clear ROI.	—

Dimension	Score	Rationale	Improvement Task
Desired AI Behaviour	3	AI role (advisory → gated autonomous) defined. Escalation boundaries exist. Safe fallback specified.	T-002: Formalise behaviour specification per autonomy level
Human–AI Workflow	3	NOC SITREP + acknowledge cycle implemented. Override tracking exists.	T-003: Implement behavioural observation feedback pipeline (§7)
Data & Knowledge	3	Multi-source strategy defined. Synthetic pipeline complete. Real data access limited.	T-004: Improve synthetic data realism (§10)
Evaluation & Metrics	3	Decision memory benchmark (0.9 threshold). Causal AI validated.	T-005: Implement continuous evaluation pipeline
Safety & Governance	2	Harm scenarios identified. Audit trail exists. **Regulatory docs are stubs.**	T-006: Write substantive OFCOM, ICO, Safety documents
Learning & Feedback	3	RLHF with multi-operator voting. Drift detection exists.	T-007: Implement structured multi-dimensional feedback (§7)

Product Readiness Score: 21/35 — Pilot Ready

Lens 2 — Enterprise AI Business Case

Dimension	Score	Rationale	Improvement Task
Operationalisation	2	Runs in development. Structured logging, health probes implemented. In-process state won't survive scale-out.	T-008: Implement persistent event bus (Redis-backed queue)
Decision Integration	3	SITREPs inform NOC decisions. Policy engine gates actions.	T-009: Prove advisory value in real deployment scenario
Risk & Accountability	2	Owners defined. Acceptable error thresholds partially established. **Regulatory docs are stubs.**	T-006 (shared): Regulatory documentation
Capability Building	2	Training curriculum drafted. NOC runbook exists. Not tested with real operators.	T-010: Validate training curriculum with pilot NOC team
Strategic Direction	3	Clear vision (intelligence fabric). Competitive moat identified (Dark Graph + Abeyance Memory).	—

Business Case Score: 12/25 — Tool Adoption

Lens 3 — Skills & Workforce Readiness

Dimension	Score	Rationale	Improvement Task
Role Transformation	2	NOC engineer supervisory role defined conceptually.	T-011: Define AI-adjusted NOC engineer role spec
Skill Acquisition	2	Training curriculum exists but untested.	T-012: Build hands-on training environment with synthetic data
Collaboration Skills	1	SITREPs are shared artefacts. No cross-team coordination protocol.	T-013: Design cross-team SITREP escalation workflow
Knowledge Integration	2	Decision Memory captures learnings. No playbook generation from patterns.	T-014: Implement automated playbook generation from high-confidence patterns
Learning Culture	1	Experimentation within dev team only.	T-015: Create "Pedk.ai Learning Hub" operator-facing knowledge base

Skills Readiness Score: 8/25 — Tool Adoption

Composite Readiness & AI Failure Pattern Mitigation

Lens	Score	Band
Product	21/35	**Pilot Ready**
Business Case	12/25	Tool Adoption
Skills	8/25	Tool Adoption

> [!WARNING] **Erdos AI Failure Pattern**: The most common failure is **Behaviour maturity > Workflow maturity > Governance maturity** — the AI works, but the product doesn't. Pedk.ai exhibits this pattern: AI capabilities are ahead of operational workflow integration and governance maturity.

Realistic mitigations the product team can take:

#	Mitigation	Targets	Priority
1	**Ship advisory-only first.** Resist the temptation to demonstrate autonomous capability. Focus every demo on SITREP quality and Dark Graph discovery.	Workflow gap	🔴 Critical
2	**Write the regulatory documents.** OFCOM pre-notification, ICO DPIA, and Safety Whitepaper must be substantive before any operator engagement.	Governance gap	🔴 Critical
3	**Implement behavioural feedback.** The highest-value learning signal is what operators *do*, not what they click. Invest here before expanding automation.	Workflow gap	🔴 Critical
4	**Build the training environment.** An untested training curriculum is not a curriculum. Build hands-on exercises using synthetic data from Sleeping-Cell-KPI-Data.	Skills gap	🟡 High
5	**Validate on reference deployment.** The committee review cycle was rigorous but internal. The next gate must involve a real NOC team.	Business case gap	🟡 High

#	Mitigation	Targets	Priority
6	**Separate "works in demo" from "works in production."** Audit every feature for production-readiness. Honest maturity assessment (§6) is the first step.	Product gap	🟡 High

15. Technology Stack

Component	Technology	Rationale
Backend API	Python 3.11+, FastAPI	ML ecosystem, async-native
Decision Store	PostgreSQL + JSONB	Flexible schema, industry standard
Time-Series	TimescaleDB (Postgres extension)	Same engine, auto-partitioning
Vector Search	pgvector	Semantic similarity for decision traces and Abeyance Memory
Streaming	Apache Kafka	Industry standard for telemetry ingestion
ML/AI	PyTorch, scikit-learn, statsmodels	Anomaly detection, causal inference
LLM	**Vendor-neutral** — supports cloud-hosted (Gemini, GPT, Claude) and local/on-prem models (LLaMA, Mistral) via abstraction layer	Explanation layer, SITREP generation. Local LLM option critical for data sovereignty.
Frontend	Next.js	NOC dashboard SPA
Containers	Docker (multi-stage, non-root)	Production-hardened
Orchestration	Kubernetes + Helm	Scalable deployment

Component	Technology	Rationale
Observability	OpenTelemetry, structured JSON logging	Distributed tracing
Testing	pytest, Locust	Functional, integration, load

> [!NOTE] **LLM vendor neutrality is a product requirement.** Pedk.ai's `LLMService` uses an abstraction layer that routes to the configured provider. Customers deploying on-premises with data sovereignty constraints can use local models. Cloud-hosted deployments can use any major provider. The product must never depend on a single LLM vendor.

16. Implementation Phases & Maturity Assessment

Foundation & Signal Fabric (Phases 1–4) ✅

✅ Multi-tenant FastAPI backend, multi-database support, ingestion pipeline, Decision Trace schema, pgvector similarity search, anomaly detection, RCA engine, LLM SITREPs, sleeping cell detection, structured logging, health probes, circuit breakers, pytest suite, Docker.

Intelligence & Standards (Phases 5–8) ✅

✅ Granger Causality, RLHF operator feedback, ADF stationarity tests, multi-operator feedback, memory optimisation, TMF642 Alarm API, TMF628 Performance API, vendor normaliser.

Enterprise Excellence (Phases 9–11) ✅

✅ Committee review cycle (3-pass + 5-member executive), JWT auth, RBAC, TLS/SSL, Kubernetes manifests, OTel tracing, NOC dashboard.

Expansion (Phases 12–14) ✅

✅ Memory benchmarking, capacity planning wedge, CX intelligence wedge.

AI Control Plane (Phase 15+) ⚠️ Prototype

Policy engine, semantic context graph, RL evaluator. BSS data layer deferred to future roadmap.

Structural Remediation (V8) 🔨 In Progress

46 tasks across 6 phases from formal architecture review.

Dark Graph V8 🔨 In Progress

Full topology reconciliation across all divergence types, with CasinoLimit/Datagerry proving ground.

17. Governance & Quality Standards

Erdos Quality Bar

Element	Condition	Status	Action
Problem Statement	Specific, measurable, owned	✅ Pass	—
Architecture	Clear layers, extensibility	✅ Pass	—
Data Pipeline	Schema contracts, governing	✅ Pass	—
Evaluation	Production monitoring, business linkage	⚠️ Partial	T-005: Continuous evaluation pipeline
Safety & Governance	Enforced controls, continuous monitoring	⚠️ Partial	T-006: Regulatory documents
Regulatory	Substantive, regulator-ready	❌ Fail	T-006: OFCOM, ICO, Safety Whitepaper
Abeyance Memory	Persistent, long-horizon, multi-modal	⚠️ Partial	T-016: Abeyance decay and cold storage

Element	Condition	Status	Action
Evidence Fusion	Pluggable, per-customer	⚠️ Partial	T-017: FusionMethodologyFactory
Operator Feedback	Behavioural + structured + quick	⚠️ Partial	T-003, T-007: Feedback pipeline
Synthetic Data Quality	Realistic naming, scenarios, temporal patterns	⚠️ Partial	T-004, T-018–T-022

Task Backlog

ID	Task	Priority	Project
T-002	Formalise AI behaviour specification per autonomy level (§7)	🟡 High	Pedk.ai
T-003	Implement behavioural observation feedback pipeline — ingest operator ITSM actions as learning signal	🔴 Critical	Pedk.ai
T-004	Improve synthetic data realism: temporal patterns, propagation delays, CMDB decay calibration	🔴 Critical	Sleeping-Cell-KPI-Data

ID	Task	Priority	Project
T-005	Implement continuous evaluation pipeline with business outcome linkage	🟡 High	Pedk.ai
T-006	Write substantive regulatory documents: OFCOM pre-notification, ICO DPIA, Safety Whitepaper, Autonomy Status Report	🔴 Critical	Pedk.ai
T-007	Implement structured multi-dimensional operator assessment (accuracy, relevance, actionability)	🟡 High	Pedk.ai
T-008	Implement persistent event bus (Redis-backed queue) to replace asyncio.Queue	🟡 High	Pedk.ai
T-009	Design and execute reference deployment scenario with advisory-only mode	🟡 High	Pedk.ai

ID	Task	Priority	Project
T-010	Validate training curriculum with pilot NOC team using synthetic data exercises	High	Pedk.ai
T-011	Define AI-adjusted NOC engineer role specification	Medium	Pedk.ai
T-012	Build hands-on training environment with Sleeping-Cell-KPI-Data	Medium	Pedk.ai
T-013	Design cross-team SITREP escalation workflow	Medium	Pedk.ai
T-014	Implement automated playbook generation from high-confidence Decision Memory patterns	Medium	Pedk.ai
T-015	Create operator-facing “Pedk.ai Learning Hub” knowledge base	Medium	Pedk.ai
T-016	Implement Abeyance Memory decay scoring and cold storage retrieval pipeline	Critical	Pedk.ai

ID	Task	Priority	Project
T-017	Implement `FusionMethodologyFactory` — pluggable evidence fusion (Noisy-OR, Dempster-Shafer)	🟡 High	Pedk.ai
T-018	Replace UUID V4 identifiers with operator-realistic human-friendly naming conventions	🔴 Critical	Sleeping-Cell-KPI-Data
T-019	Validate synthetic fault scenarios against published Tier-1 post-incident reports	🟡 High	Sleeping-Cell-KPI-Data
T-020	Implement diurnal/seasonal temporal patterns in KPI generation (beyond AR(1))	🟡 High	Sleeping-Cell-KPI-Data
T-021	Add configurable propagation delay profiles per domain boundary for cascading alarms	🟡 High	Sleeping-Cell-KPI-Data

ID	Task	Priority	Project
T-022	Calibrate CMDB degradation rates against published CMDB audit statistics	🟡 High	Sleeping-Cell-KPI-Data
T-023	Implement causal inference methodology selection: add Transfer Entropy and PCMCI alternatives	🟡 High	Pedk.ai
T-024	Wire sleeping cell detector into `main.py` scheduler (currently dead code)	🔴 Critical	Pedk.ai
T-025	Strengthen Dark Graph module: complete Divergence Report, Datagerry adapter, CasinoLimit parser	🔴 Critical	Pedk.ai
T-026	Implement Abeyance Memory multi-modal matching (structured telemetry ↔ unstructured text)	🟡 High	Pedk.ai

ID	Task	Priority	Project
T-027	Frontend decomposition: split monolithic `page.tsx` into routed pages	🟡 High	Pedk.ai
T-028	Phase 5 test suite: expand from ~5 trivial tests to comprehensive safety gate coverage	🟡 High	Pedk.ai

18. Document Catalogue

Archived (moved to `docs/archive/`)

All superseded root-level documents — see archive directory for full inventory. Key categories: - Vision documents (V3, V8, alignment reports) - Implementation roadmaps (V1–V4, agentic plans) - Executive reviews and committee assessments (6+ versions) - Phase completion summaries - Strategic audits and remediation plans - Sales and marketing drafts

Retained in Project Root

Document	Purpose
`PRODUCT_SPEC.md`	**This document** — single source of truth
`README.md`	Developer onboarding and quick start guide
`CONSTRAINTS.md`	Operational constraints
`ARCHITECTURE_S3_REMEDIATION.md.resolved`	Active remediation tracker (46 tasks, 6 phases)

Retained in `docs/`

Document	Purpose
`TASKS.md`	Active task backlog
`IMPLEMENTATION_ROADMAP_V8.md`	Active Dark Graph V8 blueprint
`ADR-001-autonomy-positioning.md`	Architecture Decision Record
`ADR-002-autonomous-execution-architecture.md`	Architecture Decision Record
`service_inventory.md`	Service module inventory
`noc_runbook.md`	NOC operations runbook
`training_curriculum.md`	Operator training curriculum
`value_methodology.md`	ROI value methodology
`shadow_mode.md`	Shadow mode deployment spec
`TELCO2_*.md`	Active Telco2 integration docs
`3PassReviewOutcome_Roadmap_V*.yaml`	Active remediation backlog (machine-readable)

Candidates for Merge

The following `docs/` files contain overlapping content and should be consolidated in a future pass:

Files	Merge Into
`confidence_methodology.md` + `ai_maturity_ladder.md`	This document (§5, §14)
`data_architecture_adr.md` + `dpia_scope.md`	`ADR-003-data-architecture.md` (new)
`amendment_status_tracker.md`	`ARCHITECTURE_S3_REMEDIATION.md.resolved` (active tracker)
`GTM_DEMO_SUITE_V8.md` + `PEDKAI_PITCH_DECK_V8.md` + `PEDKAI_V8_EXECUTION_DEEP_DIVE.md`	Single `GTM_MATERIALS.md`

Pedk.ai — AI-Native Operational Reconciliation Engine
Methodology: Erdos AI Enterprise AI Deployment Framework
Synthetic Data: Sleeping-Cell-KPI-Data Pipeline